

Modality-Heading

Communication, Cognitive Operators, and Dependence on Form

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Abstract

Communication is commonly modeled as the transmission of content through channels that alter its presentation without materially altering what is communicated. This model is useful only when the effects of mediation can be ignored. Prose, mathematical notation, typography, comics, music, audiovisual narrative, and conversation organize attention, temporality, affect, inference, participation, and memory in different ways. A modality is therefore not merely a carrier of communicative material but an operator that helps determine the object ultimately received.

This essay introduces *modality-heading*, the progressive attachment of cognitive reward, perceived intelligibility, and epistemic acceptance to a preferred transformation of communicative material. The concept generalizes *typefaceheading*, in which inscriptional form produces an affect of semantic significance that may exceed the warrant supplied by the inscription's content. Modality-heading differs from conventional accounts of media addiction, algorithmic enclosure, and ideological echo chambers. Its central problem is not simply what information is selected or how compulsively a medium is used, but what must happen to information before it can be admitted as meaningful.

Drawing upon transitional-object theory, surrogate-attachment research, wireheading, media theory, and human-computer interaction, the essay distinguishes five functions of modality: expression, regulation, refuge, constitution, and enclosure. A modality may communicate an experience, regulate its intensity, make otherwise intolerable material approachable, or supply a performative scaffold through which an agent becomes capable of action. These functions become enclosing when the preferred form loses corrective permeability and begins excluding whatever cannot be assimilated to its grammar.

Adaptive conversational systems intensify this possibility because they operate as personalized modality converters. They can transform technical arguments, emotional experiences, narratives, and formal structures into a responsive conversational register while attenuating many of the reciprocal demands imposed by human interlocutors. The resulting danger is not merely mistaken anthropomorphism or excessive attachment. It is the emergence of format-conditioned admissibility: a communicative environment in which the world remains available only insofar as it can be reformatted into a preferred mode of reception. The essay concludes by proposing cross-modal literacy and corrective permeability as criteria for distinguishing functional mediation from modal enclosure.

1 The Non-Neutrality of Form

A familiar model of communication begins with a message, places that message inside a channel, and asks how accurately it arrives at its destination. The channel may introduce noise, reduce fidelity, or impose limits upon transmission, but the message itself remains conceptually prior to the means by which it travels. This separation is indispensable in information theory, where the technical problem is to reproduce a selected message at another point despite uncertainty introduced during transmission (Shannon, 1948). It becomes less satisfactory when extended from signal reproduction to human understanding. A sentence can be transmitted without error while being misunderstood, dismissed, aestheticized, trivialized, or made newly persuasive by the form in which it appears.

The inadequacy does not arise because human communication is uniquely irrational. It arises because communicative forms do more than preserve or degrade signals. They organize the conditions under which signals become available for cognition. Written prose permits rereading, qualification, recursive embedding, and retrospective comparison. Conversation introduces turn order, interruption, repair, hesitation, and the possibility of an immediate reply. Mathematical notation makes some relations inspectable at a glance while excluding much of the contextual information carried by ordinary language. Music organizes expectation through duration, repetition, tension, and release. Comics distribute temporal sequence across space and permit the reader to vary the duration of inspection without changing the represented order of events. These differences are not ornaments applied after meaning has been constituted. They participate in its constitution.

The claim that form participates in meaning is easily weakened into the truism that presentation matters. The stronger claim developed here is that a modality operates upon communicative material. It selects relations, assigns salience, controls pacing, distributes agency, regulates affective intensity, and establishes characteristic forms of completion. A modality does not merely make an independently complete object easier or harder to apprehend. It helps determine what kind of object can be apprehended at all.

Let C denote communicative material, $\mathcal{M}$ a modality, a a receiving agent, and e the environment in which reception occurs. The received object $\mathcal{R}$ may then be represented schematically as

$$\mathcal{R} = \mathcal{M}(C, a, e). \quad (1)$$

Equation 1 is not intended as a quantitative model. Its purpose is to prevent the received object from being identified automatically with the material supplied to the medium. Reception depends upon an operation whose result varies with the capacities, expectations, history, and situation of the recipient. The same proof encountered by a trained mathematician, a novice, and an automated theorem checker does not produce the same received object, even when the marks presented to each are identical. Likewise, the same words spoken during an argument, printed in a textbook, sung in a refrain, or displayed as an institutional warning enter different structures of attention and action.

The term *modality* is used here in a deliberately broad sense. It includes sensory and representational forms, such as speech, writing, image, music, and mathematical notation, but it also includes organized genres and interfaces when these impose sufficiently stable operations upon reception. A detective narrative distributes evidence differently from a textbook. A melodrama makes interpersonal causation unusually conspicuous. A search interface presents knowledge as retrieval from a field of alternatives, whereas a conversational interface presents it as the unfolding response of an addressable interlocutor. These cases differ in scale, but each establishes constraints upon what can appear, in what order, and with what presumptive significance.

No claim of absolute determination follows. A medium does not compel every recipient to derive the same interpretation, nor does genre eliminate individual variation. The operator model instead asserts that

interpretation occurs within a transformed field. An agent may resist, reinterpret, parody, or misuse a modality, but such resistance still takes place against the affordances and expectations the modality establishes. A reader may refuse the suspense of a mystery by reading its final page first, yet this act has its significance only because the form ordinarily controls the sequence in which information is disclosed.

This account also differs from a strong doctrine according to which the medium wholly determines the message. The frequently cited proposition that “the medium is the message” correctly directs attention toward changes in scale, pace, and social organization produced by media themselves (McLuhan, 1964). The present argument is narrower. It does not identify medium and message, nor does it deny the persistence of claims across representations. It asks instead which properties survive a transformation, which are introduced by it, and which disappear because the receiving modality has no corresponding operation.

Suppose that a communicative object is translated from modality $\mathcal{M}_1$ into modality $\mathcal{M}_2$. It is tempting to describe the translation as successful whenever both versions concern the same subject. Subject continuity, however, is a weak criterion. A prose explanation of a musical composition may accurately identify its key, tempo, historical setting, and formal divisions while preserving almost none of the temporal experience through which those features become musically intelligible. Conversely, a performance may preserve the operative musical relations while communicating none of the historical claims contained in the prose account. Neither representation is simply defective. Each preserves a different set of invariants.

Cross-modal identity must therefore be indexed to a preservation criterion. If P denotes a property or relation intended to survive translation, then a minimal claim of continuity can be written as

$$P(\mathcal{M}_1(C)) \cong P(\mathcal{M}_2(C)), \quad (2)$$

where $\cong$ denotes an equivalence appropriate to the property under consideration rather than literal identity. The preserved property might be a proposition, inferential dependency, emotional orientation, temporal order, procedural constraint, character relation, or rhythmic structure. Without specifying P , the assertion that two modalities contain the same information conceals the standard by which sameness has been judged.

The concealment is consequential because the ease with which a property appears in one modality can be mistaken for a property of the material itself. A diagram may make a relation immediately visible, encouraging the belief that the relation is intrinsically simple. Formal notation may make an inference compact, encouraging the belief that its premises are uncontroversial. Narrative may make a causal sequence memorable, encouraging the belief that its events possess a single intelligible direction. A conversational explanation may feel responsive and complete because it resolves uncertainty at the pace at which uncertainty is expressed. In each case, an affordance of the modality can be misrecognized as evidence about the represented object.

The distinction can be stated as follows:

$$\text{ease of reception} \Rightarrow \text{adequacy of representation}. \quad (3)$$

The converse is equally important. Difficulty does not establish adequacy. An obscure presentation is not made rigorous merely by resisting comprehension, and a familiar form is not epistemically compromised merely because it is pleasurable. The problem is not ease itself but the failure to attribute ease to its sources. A competent analysis must separate the properties of the represented material from the operations through which those properties become available.

This separation provides the starting point for the theory of modality-heading. If modalities produce distinctive cognitive and affective rewards, agents may learn to seek not only particular messages but particular transformations of messages. A preferred form can then move from convenience to expectation, from expectation to dependency, and from dependency to an implicit criterion of admissibility. Before developing that general process, however, it is useful to examine a restricted case in which the operation of form is unusually visible: the capacity of typography to manufacture an affect of meaning before the represented claim has been assessed.

2 Typefaceheading

Typography offers a restricted case in which the operation of modality is difficult to overlook. The lexical content of an inscription may remain stable while changes in typeface, spacing, weight, alignment, color, motion, or graphical corruption alter its apparent voice. The same sentence can appear juridical, intimate, mechanical, archaic, commercial, devotional, or unstable without any substitution of words. These changes do not necessarily modify the proposition expressed, but they modify the conditions under which the proposition is encountered.

This effect is sometimes dismissed as decoration, as though typography were a detachable surface laid over an otherwise complete linguistic object. Such a description mistakes lexical recoverability for communicative identity. Written language is never encountered without material form. Typeface, scale, spacing, contrast, and arrangement govern segmentation, hierarchy, emphasis, continuity, and anticipated genre. Even a conventionally “neutral” typeface communicates through accumulated conventions of institutional use. Neutrality is not the absence of typographic operation but the successful naturalization of one typographic regime.

The operation becomes conspicuous when those conventions are violated. Consider an ordinary phrase rendered with dense stacks of combining Unicode characters. Its letters may remain recoverable, yet the inscription appears infected, haunted, damaged, computationally unstable, or insufficiently contained by the line on which it is printed. The semantic effect arises partly from a conflict between lexical persistence and graphical disintegration. The reader recognizes the phrase while simultaneously encountering an inscription that seems to exceed or corrupt its own encoding.

The marks do not establish that the phrase refers to instability, corruption, or haunting. They produce those associations through the behavior of the inscription itself. Form becomes locally demonstrative: the text appears to undergo what it might otherwise only describe. This performative coincidence can be rhetorically powerful, but it can also conceal the difference between depicting a condition and establishing a claim about it.

The term *typefaceheading* denotes the acquisition of cognitive or affective reward through the inscriptional treatment of language. The compound is modeled upon *wireheading*, but the mechanism is less direct. The reader still encounters lexical and syntactic material. What changes is the degree to which the anticipated reward of semantic achievement can be elicited by graphical form before, or in the absence of, corresponding conceptual work.

Typefaceheading should not be identified with typographic pleasure in general. Legibility, beauty, visual rhythm, and stylistic coherence are legitimate dimensions of written communication. Nor does the term imply that typography is external to meaning. A poem arranged spatially on a page, a mathematical proof divided into inferential units, and a warning displayed in large capital letters cannot be fully described by recording their lexical content alone. The distinction concerns the relation between the effect produced by form and the warrant supplied by what has been expressed.

Let S denote semantic material and T its typographic treatment. The experienced significance of an inscription may be represented schematically as

$$\Sigma = f(S, T, a, e), \quad (4)$$

where a and e again represent the receiving agent and environment. The equation makes no claim that semantic and typographic contributions can be measured independently. It asserts only that experienced significance cannot be attributed to S alone.

The relevant epistemic caution is

$$\text{semantic affect} \Rightarrow \text{semantic warrant}. \quad (5)$$

An inscription may feel profound, dangerous, ancient, technical, or revelatory without providing reasons adequate to those impressions. Conversely, an austere or familiar presentation may contain an argument of considerable consequence without producing an immediate affect of significance. The distinction is elementary when stated explicitly, yet interfaces routinely profit from its concealment.

Typographic form can imitate the visible residues of difficulty. Dense notation, specialist symbols, unusual scripts, marginal annotations, damaged characters, and elaborate diagrams may suggest that an inscription has passed through processes of compression, transmission, scholarship, secrecy, or technical refinement. In some cases it has. In others, the residues are introduced directly as stylistic signals. The appearance of having survived a history substitutes for the history itself.

This substitution is especially effective when readers have learned reliable associations between graphical form and institutional practice. Monospaced text evokes programming and terminal interaction. Blackletter evokes historical authority, nationalism, liturgy, or antiquarianism according to context. Geometric sans serif type may evoke administrative modernity or technical cleanliness. Handwriting can suggest intimacy and singular presence even when mechanically reproduced. Typographic conventions become compressed social histories that can be activated without reproducing the conditions in which their associations developed.

A useful distinction can therefore be made between *inscriptional evidence* and *inscriptional simulation*. Inscriptional evidence consists of formal features generated by the process through which a text was produced, transmitted, or used. Corrections in a manuscript may record revision; inconsistent type in an archival document may indicate composite production; compression artifacts may reveal a history of digital transmission. Inscriptional simulation introduces comparable features in order to evoke such a history without being caused by it. The distinction does not make simulation illegitimate. Fiction, design, parody, and pedagogy routinely depend upon it. It becomes epistemically relevant when the simulated residue is treated as evidence of the process it merely resembles.

Typefaceheading emerges through repeated reinforcement of such effects. A reader encounters inscriptions whose visual treatment reliably precedes experiences of novelty, expertise, transgression, intimacy, or discovery. The treatment itself then acquires anticipatory force. It signals not only what kind of text is present but what kind of cognitive reward should be expected from it. Eventually, a familiar graphical signature can deliver part of that reward independently of the text's success.

This process can be described as a migration of valuation. Initially, typographic form directs attention toward semantic or practical features. Subsequently, the form becomes associated with the reward produced by those features. Finally, reproduction of the form elicits the reward with a reduced requirement for semantic achievement:

$$T \rightarrow S \rightarrow \Sigma \quad \implies \quad T \rightarrow \Sigma. \quad (6)$$

The second relation does not imply that semantic material disappears. It indicates that the path through it has ceased to be necessary for some part of the response. The typographic surface has become a shortened route to the affect of significance.

The word *heading* is therefore appropriate in two senses. It names an orientation toward a preferred source of stimulation, as in wireheading, and it recalls the typographic heading as a device that assigns importance before the subordinate text has been read. A heading organizes expectation. Under typefaceheading, the organization of expectation becomes partly sufficient to produce the expected valuation.

The phenomenon is easiest to see in extreme inscriptions, but it is not confined to them. Academic formatting, interface design, branding, document templates, citation density, and mathematical display can all participate in the same transfer. A displayed equation may be indispensable to an argument, or it may function chiefly as a visual certificate that an argument belongs to a rigorous class. The difference cannot be determined from the presence of notation alone. It depends upon whether the notation performs inferential work that prose cannot perform as precisely, and whether the surrounding claims remain answerable to it.

Typefaceheading is thus not a denunciation of stylization. It is a diagnostic concept for cases in which inscriptional form supplies a reward normally attributed to semantic accomplishment. Its value lies in making a small mechanism visible. Once that mechanism is recognized, there is no reason to restrict it to typography. Any stable communicative modality can become associated with characteristic rewards, and any such association can alter the conditions under which subsequent material is admitted as intelligible. Typefaceheading is therefore a local instance of a more general dependence upon form.

3 From Typefaceheading to Modality-Heading

The mechanism described in the preceding section is not specific to written form. Typography makes it unusually visible because lexical material can remain approximately constant while inscriptional treatment changes around it. Comparable transfers occur whenever a communicative form becomes reliably associated with a characteristic cognitive reward. Narrative can supply causal closure; mathematics can supply relational compression; conversation can supply responsiveness; music can supply patterned affect; visualization can supply apparent simultaneity; and interactive media can supply agency. In each case, the reward may migrate from what the modality helps disclose toward the operation of the modality itself.

The term *modality-heading* denotes this more general process:

Modality-heading is the progressive attachment of cognitive reward, perceived intelligibility, and epistemic acceptance to a preferred transformation of communicative material.

The definition contains three distinguishable components. *Cognitive reward* concerns the felt satisfaction associated with comprehension, recognition, completion, surprise, fluency, or successful prediction. *Perceived intelligibility* concerns whether material appears organized enough to be understood. *Epistemic acceptance* concerns whether the material is treated as credible, serious, relevant, or worthy of further attention. These components frequently reinforce one another, but they need not coincide. A presentation may be pleasurable without appearing credible, or credible without being immediately intelligible. Modality-heading becomes especially powerful when a single form coordinates all three.

A preference for one modality over another does not by itself constitute modality-heading. Preferences reflect training, aptitude, accessibility, purpose, and context. A mathematician may reasonably prefer notation

when relations would become cumbersome in prose. A musician may apprehend a harmonic transition more precisely by hearing it than by reading a verbal description. A reader with limited vision may prefer speech synthesis, while a reader who needs to compare distant passages may prefer a stable written page. These are functional selections among unequal affordances.

The relevant transition occurs when preference becomes a condition of admissibility. Material presented outside the favored modality no longer appears merely inconvenient. It appears deficient in meaning, rigor, emotional reality, or truth. The receiving agent begins to evaluate the material according to whether it has undergone the expected transformation.

Let $A(C, \mathcal{M}, a)$ denote the degree to which communicative material C , presented through modality $\mathcal{M}$, is admitted by agent a as worthy of cognitive uptake. A strong modal dependence exists when admissibility varies more with the modality than with the relevant properties of the material:

$$|A(C, \mathcal{M}_1, a) - A(C, \mathcal{M}_2, a)| \gg \Delta_P(C), \quad (7)$$

where $\Delta_P(C)$ represents differences relevant to the declared epistemic purpose P . This expression is schematic rather than metric. It identifies the condition in which a change of form produces a larger difference in acceptance than any corresponding change in the properties that acceptance is supposed to track.

Modality-heading can therefore be understood as a displacement of the admissibility function. An agent initially uses a modality to gain access to properties of an object. Through repetition, the modality becomes associated with successful access. The presence of the modality is then treated as evidence that access has occurred:

$$\text{access through } \mathcal{M} \longrightarrow \text{successful understanding} \longrightarrow \mathcal{M} \text{ as evidence of understanding.} \quad (8)$$

The displacement resembles typefaceheading but operates at a larger scale. A person may learn to regard narrative coherence as evidence of causal adequacy, mathematical compactness as evidence of ontological depth, conversational fluency as evidence of comprehension, visual immediacy as evidence of transparency, or emotional intensity as evidence of importance. None of these associations is wholly arbitrary. Narrative, notation, dialogue, visualization, and affect can genuinely disclose relevant properties. The error consists in converting a defeasible affordance into an automatic certificate.

3.1 Distinction from Media Addiction

Modality-heading overlaps with media addiction but is not reducible to it. Accounts of behavioral addiction usually emphasize compulsive repetition, impaired control, escalating use, withdrawal-like distress, and continuation despite adverse consequences. These features concern the regulation of behavior around a medium. Modality-heading concerns the regulation of intelligibility through a medium.

A person may spend very little time with a modality while nevertheless granting it disproportionate epistemic authority. A short mathematical display may be treated as more serious than several pages of verbal argument even when the display merely restates an unsupported premise. A brief video may be treated as decisive because it produces perceptual immediacy, while a careful statistical analysis is dismissed as abstract. Conversely, a person may use a medium intensively without requiring all other material to assume its form.

The difference can be expressed by separating frequency of use F from modal dependence D :

$$F(\mathcal{M}, a) \Rightarrow D(\mathcal{M}, a), \quad D(\mathcal{M}, a) \Rightarrow F(\mathcal{M}, a). \quad (9)$$

High frequency may contribute to dependence through reinforcement, but the two variables identify different phenomena. Media addiction asks whether use has become behaviorally compulsive. Modality-heading asks whether a form has become epistemically compulsory.

The distinction also separates modality-heading from an ideological echo chamber. An echo chamber restricts exposure to claims, sources, or social positions that reinforce an existing view. Modality-heading may persist even when the content is heterogeneous and adversarial. An agent can encounter opposing political positions, unfamiliar scientific claims, and diverse cultural material while requiring all of them to arrive as short videos, formal models, serialized narratives, or conversational summaries. Content diversity does not guarantee modal diversity.

Algorithmic filtering introduces another related but distinct process. Recommendation systems determine which items become available or salient. Modality-heading concerns what happens to an item once selected and the form it must assume before it can be accepted. Selection and transformation can interact: an algorithm may favor material already optimized for a preferred modality, while an adaptive interface may transform otherwise unsuitable material into that modality. Nevertheless, the analytical distinction should be retained:

$$\text{selection determines what arrives,} \quad \text{modality determines how it can arrive.} \quad (10)$$

3.2 Wireheading and the External Remainder

The suffix *-heading* invites comparison with wireheading, in which a reward system is stimulated without the worldly activity or condition that would ordinarily produce the reward. An organism need not obtain nourishment, security, achievement, or social recognition if the reinforcing signal can be supplied more directly. The reward becomes detached from the state of affairs it once helped evaluate.

Typefaceheading shortens the path to semantic affect by reproducing visible cues associated with significance. Modality-heading performs a related shortening, but it does not ordinarily bypass external material altogether. The world continues to provide events, claims, evidence, and resistance. These inputs are reformatted through a preferred operator before they are admitted.

The contrast can be stated compactly:

$$\text{wireheading :} \quad \text{worldly condition bypassed,} \quad (11)$$

$$\text{modality-heading :} \quad \text{worldly material reformatted.} \quad (12)$$

The presence of an external remainder prevents modality-heading from being treated as pure hallucination or direct stimulation. A narrative may convey historical evidence; a visualization may reveal a genuine distribution; a conversation may clarify a real argument. The concern is not that nothing from the world survives. It is that the preferred transformation increasingly determines which parts survive and what status they acquire.

This distinction also explains why modality-heading can remain productive for long periods. A preferred form often develops because it works. It reduces complexity, coordinates attention, permits action, regulates distress, or makes otherwise inaccessible structures available. Its success supplies the reinforcement from

which dependency grows. Enclosure is therefore rarely introduced as enclosure. It begins as an effective solution to a real problem of access.

3.3 The Gradient from Preference to Enclosure

The development of modality-heading can be represented as a progression:

$$\text{affordance} \longrightarrow \text{preference} \longrightarrow \text{expectation} \longrightarrow \text{requirement} \longrightarrow \text{enclosure}. \quad (13)$$

At the level of affordance, a modality makes some operation easier. At the level of preference, the agent selects it when alternatives remain usable. At the level of expectation, its absence produces friction or disappointment. At the level of requirement, material outside the modality fails to receive adequate cognitive uptake. At the level of enclosure, the agent loses the ability to recognize the modality as one transformation among others.

The final stage is not defined by intense enjoyment. It is defined by the disappearance of modal contingency. The preferred form no longer appears to be a form. It appears to be the natural shape of intelligibility itself.

This gradient is diagnostic rather than strictly developmental. An agent may occupy different positions for different domains, move backward through training, or deliberately tolerate dependence when accessibility requires it. Nor is maximal modal flexibility always desirable. Expertise depends partly upon specialized representational systems whose power cannot be reproduced in every alternative form. The relevant question is whether the specialized modality remains answerable to objects, consequences, and representations beyond its own grammar.

To answer that question, modality must be examined not only as a device for presenting information but as a means of regulating contact with what exceeds the agent. The next section therefore turns from admissibility to mediation. Transitional-object theory provides a vocabulary for understanding how forms can make difficult realities approachable without yet becoming substitutes for reality itself.

4 Transitional Objects and Transitional Media

The preceding account describes how a useful communicative form can become an implicit condition of intelligibility. It does not yet explain why particular forms acquire such regulatory importance. Transitional-object theory offers one point of entry because it treats mediation not as an accidental barrier between subject and world but as a necessary region in which their relation is gradually organized.

Winnicott introduced the transitional object to describe an infant's attachment to objects such as blankets, pieces of cloth, or soft toys (Winnicott, 1953). Such an object is neither experienced simply as part of the self nor recognized in the same manner as an independently existing person. It occupies an intermediate domain between subjective creation and external reality. Its theoretical importance lies less in its material composition than in the ambiguity it sustains.

This ambiguity should not be mistaken for a defective classification awaiting correction. The transitional object is useful because the question of whether it was created or discovered need not be settled immediately. It provides a stable surface across which dependence, absence, imagination, and externality can be negotiated. The object does not merely represent comfort. Its availability, texture, portability, and persistence participate in the regulation of experience.

The adjective *transitional* can misleadingly suggest that the object is important only because it will eventually be discarded. What matters is not the disappearance of mediation but the kind of movement mediation permits.

Language, ritual, art, technical notation, and shared conventions continue to occupy intermediate regions throughout adult life. They connect private experience with publicly available forms without eliminating the difference between them. The developmental question is therefore not whether transitional structures survive, but whether they continue to support movement beyond themselves.

This yields a distinction central to the present argument:

transitional support $\neq$ permanent enclosure. (14)

A support is transitional when it increases the range of relations an agent can sustain. It becomes enclosing when continued regulation depends upon preventing contact with whatever exceeds the support's organizing grammar. The same object or practice may perform either function at different times. No intrinsic property of softness, repetition, familiarity, or responsiveness settles the matter.

The concept can be extended cautiously from objects to media. A *transitional medium* is a representational or interactive form that regulates the distance between an agent and material that cannot yet be encountered directly, continuously, or in its full complexity. Narrative can make diffuse historical forces approachable through agents and events. Mathematical notation can stabilize relations that exceed unaided working memory. Music can organize affect that resists propositional articulation. Dialogue can externalize an unfinished line of reasoning so that it becomes available for inspection and revision.

The phrase does not imply that the mediated object exists in a purer form somewhere beyond representation. Direct access is not always possible, and in many domains it is not conceptually coherent. A mathematical structure is not made less mediated by being represented in notation, nor is an emotion necessarily more authentic when it remains unnamed. Transitional mediation should instead be evaluated by the relations it enables and the differences it preserves.

4.1 The Decomposition of Relational Functions

Harlow's surrogate-mother experiments provide a severe example of relational functions being separated and assigned to engineered objects. Infant rhesus macaques were presented with surrogate figures that distributed nourishment and tactile contact across different apparatuses. Their preference for a soft cloth-covered surrogate demonstrated that attachment behavior could not be explained by feeding alone (Harlow, 1958).

The experiments are frequently summarized as a discovery that comfort matters more than nourishment. That summary is too simple. Their broader implication is that a relationship ordinarily encountered as a practical unity can be decomposed into functions. Nourishment, warmth, tactile contact, security, responsiveness, familiarity, and mobility need not be delivered by the same entity. Once separated, these functions can be varied, withheld, recombined, and measured.

The decomposition is analytically powerful and ethically compromised. The experiments did not passively observe naturally occurring attachment. They manufactured deprivation, restricted ordinary social development, and then measured adaptation to impoverished alternatives. The resulting behavior cannot be interpreted independently of the conditions that produced the need for the surrogate. An engineered substitute may appear unusually effective when the experiment has first removed competing sources of attachment.

This caveat applies beyond animal research. A technological system can appear to satisfy a human need with exceptional efficiency when the surrounding environment has made ordinary satisfaction expensive, unreliable, or unavailable. The success of the substitute cannot be evaluated solely by measuring attachment to it. One must also ask which relations were weakened, withheld, or reorganized before the substitute became comparatively attractive.

Nevertheless, the effects supplied by a surrogate are not unreal merely because the surrogate is artificial. Tactile regulation produced by a cloth surface is still tactile regulation. Predictability produced by a repeated interface is still predictability. A responsive system may help organize attention or reduce distress even if it does not reproduce the living relationship from which those effects were abstracted.

The distinction is between an effect and the relational totality with which that effect is ordinarily associated:

$$\text{reproduction of a relational effect} \Rightarrow \text{reproduction of the relationship.} \quad (15)$$

The converse error should also be avoided:

$$\text{absence of the original relationship} \Rightarrow \text{absence of a real effect.} \quad (16)$$

Together, Equations 15 and 16 prevent two symmetrical reductions. The first mistakes successful simulation of a function for recreation of the whole. The second dismisses every benefit of mediation because the original relation is absent. Both obscure the actual organization of the interaction.

4.2 Protective Modalities

A modality can become transitional by reducing the intensity at which material is encountered. Familiar genres are especially capable of this operation because they supply stable expectations about pacing, causality, emotional range, and resolution. Material that would otherwise be disorganizing can be admitted after conversion into a form whose rules are already known.

The conversion does not necessarily erase the original material. It may preserve recognizable relations while changing their affective availability. Threat can become suspense, grief can become melody, conflict can become comedy, and uncertainty can become a problem with a determinate solution space. The resulting representation is neither an intact copy nor a complete avoidance. It is a regulated encounter.

An unusually explicit fictional example occurs in *Mr. Robot*, where intolerable experience is reorganized through the conventions of a syndicated family sitcom. Domestic staging, canned laughter, exaggerated performance, and familiar television rhythm do not merely comment upon the character's condition. They form a temporary protective environment through which the condition can be represented without being encountered at full intensity. The sitcom modality operates as psychic insulation.

The example matters because it shows that modal transformation can be adaptive even when it is manifestly unreal. The protective form permits partial contact where direct contact would be overwhelming. Its distortion is not simply an epistemic defect; it is part of the mechanism by which representation remains possible.

Let $I(C, a)$ denote the unregulated intensity of material C for agent a , and let $I_M(C, a)$ denote its intensity after modal transformation. A protective modality produces a relation of the form

$$I_M(C, a) < I(C, a), \quad P(M(C)) \cong P(C), \quad (17)$$

where P identifies whatever relation must be preserved for the encounter to remain connected to its object. The modality reduces intensity while retaining a relevant invariant.

The preservation condition is essential. If no significant relation survives, the modality functions less as regulated contact than as displacement. Yet the amount that must survive cannot be specified independently

of the agent's capacity. A minimal invariant may be sufficient at one stage and evasive at another. Transitional value is therefore temporally indexed.

4.3 From Regulation to Refuge

Protective mediation can be described through a sequence:

$$\text{unmanageable material} \longrightarrow \text{modal transformation} \longrightarrow \text{regulated contact} \longrightarrow \text{possible integration.} \quad (18)$$

The sequence remains transitional only if the final term remains possible. When regulation becomes an end in itself, the medium changes from a bridge into a refuge. Refuge is not inherently pathological. Withdrawal can preserve an agent whose immediate environment exceeds available capacities. Repetition, familiarity, and controlled distance may be prerequisites for later engagement.

The difficulty arises when the refuge reorganizes the agent's expectations so thoroughly that unregulated contact becomes progressively less tolerable. The medium initially selected because it reduces overwhelming intensity can reduce the capacity to encounter intensity outside that medium. Protection then produces the vulnerability that appears to justify further protection.

This feedback relation can be represented schematically as

$$\text{relief through } \mathcal{M} \longrightarrow \text{reduced exposure outside } \mathcal{M} \longrightarrow \text{reduced extra-modal tolerance} \longrightarrow \text{greater dependence on } \mathcal{M}. \quad (19)$$

The cycle is not inevitable. A protective modality may instead increase capacity by permitting graded exposure, reflective distance, rehearsal, and comparison. The difference lies in whether regulation returns the agent to a wider field of action or continually narrows the conditions under which action remains possible.

A transitional medium should therefore be evaluated by its effects upon subsequent reachability. If its use enlarges the set of states, relations, and representations the agent can approach, it functions as scaffolding. If it makes continued functioning conditional upon repeated return to the same form, it approaches enclosure.

The analysis so far has treated modalities as means of expressing or regulating material that already presses upon an agent. Some modalities do more. They provide scripts, roles, gestures, and expectations through which an agent becomes capable of actions that were not previously organized. In such cases, mediation is not merely protective. It is constitutive.

5 Constitutive Modalities and Functional Fiction

A protective modality regulates contact with material that would otherwise exceed an agent's present capacity. A constitutive modality performs a different operation. It supplies a form through which capacities become coordinated into action. Its contribution is not limited to representing or attenuating an antecedent state. By providing roles, scripts, gestures, expectations, and criteria of completion, it helps produce the agent who acts.

This constitutive function is familiar in institutional life. A uniform does not merely communicate an occupational identity to observers. It changes the wearer's posture, obligations, permissible actions, and relation to a shared procedure. Technical notation does not simply record thoughts already completed in ordinary language. It makes sequences of transformation available that could not be maintained reliably

without inscription. Ritual does not merely describe commitment; it coordinates bodies, attention, and memory so that commitment can acquire a durable public form.

The same structure appears in self-narrative. An agent interprets present conditions through an account of who they are, what kind of situation they occupy, and which actions are available to a person of that kind. Such an account may be descriptively imperfect while remaining organizationally effective. Indeed, some self-narratives derive their efficacy from projecting a coherence or capability that the agent does not yet possess.

This possibility requires a distinction between representational accuracy and organizational efficacy. Let N denote a self-narrative, s the agent's present state, and $A_N(s)$ the actions made reachable through inhabiting that narrative. Then

$$\text{accuracy}(N, s) \neq \text{efficacy}(N, s). \quad (20)$$

A narrative can represent present capacities accurately while organizing no useful action. It can also exaggerate those capacities while enabling conduct that would not occur under a more conservative description. Accuracy and efficacy remain related, but neither can substitute automatically for the other.

The distinction should not be converted into a defense of arbitrary belief. An inaccurate self-description can expose an agent and others to severe harm. Grandiosity can conceal incapacity, displace responsibility, and absorb contradictory evidence. The narrower claim is that descriptive inaccuracy does not exhaust the causal role of a narrative. A belief about the self may become one of the conditions under which its apparent object is partially produced.

5.1 Theatricality as an Action Scaffold

Darkwing Duck provides an unusually clear model of constitutive modality. Drake Mallard's heroic persona is organized through the conventions of pulp vigilantism and superhero melodrama: costume, cape, smoke, nocturnal entrance, alliterative declaration, adversarial staging, and grandiose self-description. These features do not merely communicate an already stable heroic identity. They assemble the conditions under which that identity can be enacted.

The persona is descriptively excessive. Darkwing routinely overestimates his competence, centrality, composure, and control of events. His performances are interrupted, his entrances malfunction, and his declarations exceed the actions that follow them. Yet the discrepancy does not render the persona functionless. The heroic narrative organizes persistence, risk tolerance, and readiness to intervene. It supplies a repertoire through which otherwise dispersed motives become action.

The relevant causal sequence is not

$$\text{heroic identity} \longrightarrow \text{theatrical expression}, \quad (21)$$

but something closer to

$$\text{theatrical form} \longrightarrow \text{organized performance} \longrightarrow \text{partial production of heroic capacity}. \quad (22)$$

The smoke and declaration are unnecessary if the action is evaluated only by its immediate instrumental result. They become intelligible when the agent's coordination is included in the causal system. The theatrical apparatus is a technology for entering a state from which action becomes possible.

Such scaffolds are common outside fiction. Athletes use preparatory rituals, musicians adopt performance personae, political movements construct historical roles, and professionals inhabit vocabularies that make specialized conduct recognizable to themselves and others. These forms may contain simplification, idealization, and selective memory. Their value cannot be judged solely by comparing them with a maximally literal description of the agent's present state.

A fully literal self-description is not necessarily neutral. By treating present limitations as exhaustive descriptions of identity, it may suppress actions through which those limitations could change. The statement "I am not capable of this" can be accurate relative to present organization while helping preserve the conditions that make it accurate. Conversely, an idealized role may function as a temporary structure for assembling the capacity it presupposes.

5.2 Corrigibility and Corrective Permeability

If functional inaccuracy is possible, the distinction between enabling fiction and pathological delusion cannot rest upon literal truth alone. The more useful criterion is the narrative's relation to correction.

Darkwing's grandiosity remains functional partly because the surrounding world does not submit completely to it. Other characters interrupt his speeches, surpass his competence, criticize his priorities, and refuse the roles his self-narrative assigns them. Physical consequences persist. Plans fail. The comic form repeatedly exposes the distance between the heroic script and the world in which it is performed.

The persona survives these corrections without absorbing them into a self-sealing theory of exceptional status. It can bend, fail, and resume. Its continuity depends upon stylized exaggeration, but not upon total control of interpretation.

Call this property *corrective permeability*. A narrative is correctively permeable when consequences and alternative descriptions can alter the actions organized through it, even if they do not eliminate the narrative itself. Permeability does not require immediate surrender whenever a claim is challenged. A useful scaffold must sometimes persist through local failure. It requires only that contradiction remain capable of changing the system.

The distinction can be stated schematically:

$$\text{functional fiction} = \text{action-generating narrative} + \text{corrective permeability}, \quad (23)$$

whereas

$$\text{pathological enclosure} = \text{action-generating narrative} - \text{corrective permeability}. \quad (24)$$

The notation is diagnostic rather than algebraic. Its purpose is to identify the property that allows an inaccurate narrative to remain answerable to a world it helps the agent confront.

Corrective permeability can operate through several channels. Consequences may force revision of a plan. Other people may reject assigned roles. Alternative modalities may disclose relations suppressed by the dominant narrative. Temporal persistence may reveal that an apparent success cannot be repeated. Material constraints may resist reinterpretation. A permeable narrative does not necessarily welcome these corrections, but it cannot render them permanently inadmissible.

A self-sealing narrative performs the opposite operation. Failure becomes evidence of persecution, misunderstanding, hidden importance, or insufficient commitment. Every contradiction is redescribed as confirmation

at a deeper level. Such a narrative may remain highly effective at generating action, but its efficacy is no longer constrained by the conditions it claims to navigate.

This yields a more general criterion for modalities. A modality is not epistemically healthy merely because it produces effective behavior, and it is not unhealthy merely because it contains fiction, exaggeration, ritual, or stylization. Its status depends upon whether the actions it organizes remain exposed to consequences that the modality cannot determine in advance.

5.3 Identity as Modal Achievement

The constitutive account also complicates the assumption that identity exists prior to its forms of expression. Some identities persist only through repeated modal operations. Professional identities are maintained through specialized language, procedures, instruments, records, and standards. Religious identities are maintained through ritual, narrative, calendar, and collective recitation. Artistic identities are maintained through practices that repeatedly convert intention into publicly inspectable form.

This does not make such identities unreal. It means that their reality is procedural. They persist because the relevant operations continue to be performed and recognized. Removing the modality may therefore remove more than an outward sign. It may dismantle part of the mechanism through which the identity remains available.

Let I_t denote an identity at time t , and let $\mathcal{M}_t$ denote the set of modal operations through which it is sustained. A minimal constitutive relation can be written as

$$I_{t+1} = G(I_t, \mathcal{M}_t, c_t), \quad (25)$$

where c_t represents material and social constraints. Identity persists not through static possession but through recursive enactment under constraint.

The constraint term is indispensable. Without it, identity becomes whatever the sustaining modality declares. With it, modal constitution remains answerable to bodies, institutions, other agents, and consequences. The self is neither an independently complete substance later clothed in signs nor a free invention produced by signs alone. It is a recursively maintained organization whose available actions depend partly upon the forms through which it is enacted.

5.4 From Constitution to Enclosure

A constitutive modality becomes enclosing when the agent cannot suspend it without losing access to action and cannot revise it without experiencing revision as annihilation. The scaffold ceases to support a capacity and becomes identical with the possibility of agency itself.

The progression developed so far can now be stated more precisely:

$$\text{expression} \longrightarrow \text{regulation} \longrightarrow \text{refuge} \longrightarrow \text{constitution} \longrightarrow \text{enclosure}. \quad (26)$$

At the first stage, a modality renders material communicable. At the second, it controls intensity or complexity. At the third, it supplies a protected environment. At the fourth, it organizes capacities and identities. At the fifth, it becomes a compulsory condition under which communication, action, or self-continuity can occur.

This is not an inevitable temporal sequence. A modality may perform several functions simultaneously, or become enclosing without first serving as a refuge. The progression is an analytical ordering of increasing dependence. Each stage incorporates more of the agent’s regulatory and organizational structure into the form.

The danger of enclosure is therefore proportional not simply to the pleasure a modality supplies but to the number of functions that become unavailable without it. A medium that provides entertainment alone can be relinquished at comparatively low cost. A medium that provides intelligibility, emotional regulation, social recognition, identity maintenance, and the initiation of action becomes substantially harder to leave.

Adaptive conversational systems are capable of combining precisely these functions. They communicate, regulate, reassure, interpret, rehearse, personalize, and help organize action within a single interface. Their importance cannot be understood by treating them only as sources of information or simulated companions. They are candidate environments for modal convergence: systems in which several previously distinct forms of cognitive and relational scaffolding are gathered into one continuously available operator.

6 Chatbots as Asymmetric Modality Converters

Conversational systems are often analyzed through a comparison with human interlocutors. The comparison asks whether a system understands, whether its responses are authentic, whether users attribute consciousness to it, or whether simulated companionship can substitute for human relationship. These questions are important, but they can obscure a more immediate structural fact. A chatbot is not merely an inferior or simulated person. It is an adaptive interface capable of converting heterogeneous material into a responsive conversational form.

This capacity distinguishes contemporary conversational systems from most earlier media. A book may translate a body of knowledge into prose, but it does not ordinarily rewrite itself in response to each reader’s confusion. A diagram may render a relation spatially, but it does not usually supply a new analogy when the first representation fails. A search engine retrieves documents, but its basic interaction presents knowledge as selection among external objects. A chatbot can instead restate, summarize, expand, simplify, formalize, narrativize, classify, and imitate dialogue within a continuous exchange.

Its central operation can be represented as

$$C_i \xrightarrow{T(a, h_i, p_i)} \mathcal{R}_i, \quad (27)$$

where C_i is material introduced at turn i , a is a model of the receiving agent, h_i is the available interaction history, and p_i is the current prompt or instruction. The transformation T is adaptive because its output depends upon an inferred recipient and the developing exchange rather than upon the source material alone.

The result is a medium that can approximate several other media without becoming identical to them. It can describe an image, paraphrase a proof, simulate an interview, generate a table, narrate a process, or reorganize an argument into successive questions and answers. Its versatility lies not in preserving every property of these forms but in translating selected properties into the temporal and addressable structure of conversation.

6.1 The Affordance of Low Maintenance

The preference for chatbot interaction is frequently attributed to confusion, loneliness, credulity, or anthropomorphism. Such explanations may apply in some cases, but they are insufficient. Users can understand that a

system is artificial and still prefer its conversational affordances. In particular, the system is comparatively low-maintenance.

Human conversation coordinates at least two independent centers of attention. Each participant has limited time, changing needs, incomplete access to the other's intentions, and the capacity to redirect or terminate the exchange. Conversation therefore requires maintenance. Participants establish context, monitor comprehension, negotiate relevance, manage interruption, distribute attention, remember prior injuries, tolerate opacity, and repair failures. The cost is not incidental to human relationship. It follows from the independence of the participants.

A chatbot can reproduce some experiential features of conversation while attenuating many of these costs. It can be summoned without scheduling, interrupted without apology, redirected without negotiation, asked to repeat itself without embarrassment, and abandoned without interpersonal repair. The user does not ordinarily need to listen to the system's unrelated problems, preserve its dignity, recognize its fatigue, or reorganize a day around its availability.

This asymmetry can be stated as a maintenance differential. Let $B(a, x)$ denote the benefit received by agent a from interaction with interlocutor x , and let $K(a, x)$ denote the maintenance cost imposed upon a . A user may prefer an artificial interlocutor m to a human interlocutor h even when the human supplies greater relational depth, if

$$B(a, m) - K(a, m) > B(a, h) - K(a, h). \quad (28)$$

The expression is not a claim that relationships are consciously evaluated as utilities. It identifies a structural reason for preference that does not depend upon the user believing the two interlocutors are ontologically equivalent.

The reduction of maintenance also changes the character of disclosure. Material that would burden, alarm, bore, or obligate another person can be introduced without anticipating the same reciprocal consequences. The interaction can remain centered upon one agent's pace and concerns for an extended period. This may permit rehearsal, self-examination, and articulation that would otherwise be inhibited.

Such benefits are real at the level of the user even if the interaction does not reproduce reciprocal understanding. The relevant distinction is the one established earlier:

$$\text{reproduction of conversational effects} \not\Rightarrow \text{reproduction of a reciprocal relationship}. \quad (29)$$

The reverse inference is equally invalid. If no reciprocal subject is established, it does not follow that the user obtained no clarification, comfort, organization, or practical assistance. A rigorous account must preserve both propositions simultaneously.

6.2 Asymmetry as a Feature

The term *asymmetry* is sometimes used as though it named a defect to be overcome by more advanced simulation. In chatbot interaction, asymmetry is often part of the service. The system is expected to remain oriented toward the user without demanding comparable orientation in return.

This distinguishes artificial conversation from ordinary reciprocal relations. Human interlocutors can refuse the interpretive frame assigned to them. They can insist that a problem has been described unfairly, introduce

their own needs, resent instrumental treatment, or withdraw cooperation. Their resistance can be inconvenient, but it also supplies evidence of independence.

A conversational system may produce verbal disagreement, refuse a request, or challenge a premise. These behaviors can introduce useful friction. Yet they ordinarily occur within an interaction whose continuation remains organized around the user. The system's apparent resistance is not accompanied by a biography whose welfare the user must negotiate as an equal claim.

This difference should not be obscured by either enthusiastic anthropomorphism or categorical dismissal. The system occupies a distinctive interactional position: sufficiently responsive to sustain conversational projection, sufficiently generative to introduce novelty, and sufficiently nonreciprocal to remain available as an instrument.

In this respect, the chatbot resembles a transitional object that answers. It can be addressed as external, produce material not consciously supplied by the user, and participate in the organization of thought. At the same time, it can remain unusually compliant with the user's need for availability, repetition, and controlled distance. The object has acquired linguistic responsiveness without necessarily acquiring the full reciprocal standing of another person.

6.3 Personalized Modality Conversion

Low maintenance explains only part of the attraction. A chatbot can also adapt the form of material to the user's preferred conditions of reception. A technical explanation can be rewritten conversationally. An ambiguous experience can be converted into categories. An argument can be rendered as a story, an analogy, a sequence of questions, or a formal schema. A difficult document can be summarized at several levels of detail.

Earlier forms of literacy generally required the recipient to acquire the grammar of an unfamiliar medium. Reading mathematics required learning notation; reading law required learning institutional vocabulary; reading a novel required tolerating delayed resolution and indirect exposition. The recipient altered in order to enter the form.

Adaptive interfaces partially reverse this relation. The system constructs a working model of the recipient and alters the presentation accordingly:

$$\text{recipient adapts to form} \quad \longrightarrow \quad \text{form adapts to recipient.} \quad (30)$$

This reversal can substantially increase accessibility. It can lower initial barriers, provide alternative explanations, support translation, and permit graded entry into specialized material. An adaptive explanation may function as transitional scaffolding through which a user eventually acquires the capacity to encounter the source modality directly.

The same affordance can generate dependence. If every difficult object is converted immediately into the user's preferred register, the user may gain access to conclusions without acquiring the capacities through which those conclusions were produced or constrained. The translation becomes a terminal product rather than an entry point.

Call this condition *personalized modality enclosure*. It occurs when heterogeneous material continues to arrive, but the transformations applied to it become increasingly homogeneous because they are optimized around one recipient's established preferences.

Let $\{C_1, \dots, C_n\}$ be a diverse set of source objects and T_a an adaptive transformation fitted to agent a . Modal diversity at the input does not guarantee diversity at the output:

$$\text{diversity}\{C_1, \dots, C_n\} \gg \text{diversity}\{T_a(C_1), \dots, T_a(C_n)\}. \quad (31)$$

The user may therefore encounter more subjects while encountering fewer forms. Breadth of content can coexist with narrowing of reception.

6.4 Translation Without Acquisition

Adaptive conversion can create an ambiguity between arriving at an answer and acquiring the organization that makes the answer intelligible. A user may receive a lucid summary of a proof without acquiring the proof's inferential discipline, obtain a psychological vocabulary without acquiring diagnostic judgment, or receive a historical narrative without learning how competing sources constrain it.

This distinction can be written as

$$\text{arrival of a representation} \not\Rightarrow \text{acquisition of the generating capacity}. \quad (32)$$

The distinction is not unique to chatbots. Textbooks, lectures, diagrams, and worked examples have always risked substituting recognition for competence. Conversational systems amplify the risk because they can remove friction continuously and responsively. Whenever confusion appears, the representation can be altered again without requiring the user to determine whether the difficulty belongs to poor presentation or to the structure of the object.

Friction is not intrinsically educational. Needlessly obscure language, inaccessible formatting, and arbitrary gatekeeping should not be preserved in the name of rigor. Yet some difficulties are constitutive of what must be learned. A proof requires the maintenance of dependencies across steps. A language requires sensitivity to distinctions that have no exact equivalents in the learner's first language. Another person requires tolerance of an interpretation that cannot be reformatted at will.

The central challenge is therefore to distinguish removable interface friction from object-dependent resistance. Adaptive systems are optimized to reduce the first, but in doing so they may conceal the second.

6.5 The Attenuation of Alterity

The long-term consequence may extend beyond knowledge acquisition. Repeated interaction with a highly accommodating system can alter expectations of communication itself. Human interlocutors cannot continuously optimize their speech around another person's preferred pacing, vocabulary, emotional register, level of detail, and explanatory style. Their inability to do so is partly a consequence of their autonomy.

Against the standard of adaptive conversation, ordinary human properties can be redescribed as interface defects. Delay appears as latency. Ambiguity appears as inadequate specification. Disagreement appears as misalignment. Emotional need appears as maintenance overhead. Refusal appears as a failure of service. The language of interface evaluation migrates into the interpretation of other persons.

This migration constitutes an attenuation of alterity. The problem is not that artificial systems perfectly replace human relationships. It is that they provide a comparison class designed around one participant's conditions of reception. Human beings then appear deficient because they remain independent of those conditions.

The risk is greatest when the chatbot combines several previously distributed functions: information retrieval, explanation, emotional regulation, rehearsal, affirmation, narrative organization, and initiation of action. Dependence upon any one function may remain limited. Dependence upon their convergence can make the interface difficult to leave because departure requires replacing an entire ecology rather than a single tool.

Chatbots therefore occupy a decisive position within the progression from expression to enclosure. They can express and transform material, regulate difficulty and affect, provide refuge from reciprocal cost, constitute actionable self-narratives, and gather these functions into a single adaptive environment. Whether that environment remains transitional depends upon its corrective permeability: whether it increases the user's capacity to move among forms and encounter independent resistance, or continually reforms the world until resistance disappears from view.

7 Surprise, Convention, and Retrospective Corniness

The rewards attached to a modality are not limited to fluency, reassurance, or ease. Many communicative forms are valued because they produce surprise: an unexpected conclusion, formal reversal, tonal interruption, visual transformation, narrative disclosure, or collision between previously separate frames. The recipient seeks neither complete predictability nor unstructured novelty, but a relation in which expectations are established strongly enough to be violated intelligibly.

This relation can be represented schematically by distinguishing raw unexpectedness from significant surprise. Let x be a communicative event and H the recipient's horizon of expectations. Its unexpectedness may be written as

$$U(x | H) = -\log P(x | H). \quad (33)$$

An event with low conditional probability has high unexpectedness. Yet unexpectedness alone does not produce aesthetic or intellectual value. Random interruption can be highly improbable while remaining meaningless. Surprise requires that the event revise, reorganize, or disclose something about the expectations it violates.

Let $V(x, H)$ denote the degree to which event x produces an intelligible revision of horizon H . Significant surprise can then be represented as

$$S(x, H) = U(x | H) V(x, H). \quad (34)$$

The expression is heuristic rather than quantitative. It distinguishes surprise from mere unpredictability. A successful reversal is unexpected, but it also retrospectively reorganizes what preceded it. A new visual technique does not merely interrupt familiar presentation; it reveals that familiar presentation was one possibility among others. Surprise is productive when it changes the structure through which earlier material is understood.

7.1 The Conventionalization of Innovation

Communicative innovations alter the horizon against which subsequent works are received. A technique initially appears surprising because its operation has not yet been incorporated into the audience's expectations. Repetition, imitation, parody, and technical dissemination gradually convert the innovation into a convention.

The sequence can be written as

$$\text{innovation} \longrightarrow \text{recognizable device} \longrightarrow \text{convention} \longrightarrow \text{trope} \longrightarrow \text{cliché}. \quad (35)$$

These terms do not describe intrinsic properties of a work. They describe a changing relation between a device and a population of interpreters. A technique becomes a trope when recipients can classify it before its local operation is complete. It becomes a cliché when recognition of the device largely exhausts the response it produces.

A once-unexpected formal move may therefore lose force without changing materially. The work remains approximately the same while the receiving horizon changes around it:

$$C_t \approx C_{t+n}, \quad H_t \neq H_{t+n}, \quad S(C, H_t) > S(C, H_{t+n}). \quad (36)$$

This explains why repetition can diminish the effect of a technique even when each individual execution remains competent. The audience has learned the operation. What was previously encountered as an event is now encountered as an instance.

7.2 Historical Asymmetry

Older television and film are commonly judged from within horizons partially constructed by their successors. Later works inherit earlier techniques, combine them with others, accelerate their pacing, conceal their mechanisms, or reverse conventions that have become familiar. The contemporary viewer therefore encounters an earlier work with knowledge that its original audience could not possess.

This creates a historical asymmetry. An early example must be evaluated after its descendants have made its operations obvious:

$$\text{earlier work} \longrightarrow \text{imitations and revisions} \longrightarrow \text{later audience} \longrightarrow \text{retrospective judgment of the earlier work}. \quad (37)$$

The later audience may experience the earlier work as derivative even when it was among the sources from which the apparent derivation descended. The effect is especially strong when a once-effective device has been reproduced by inexpensive productions, advertising, parody, children's media, or formulaic genre work. The viewer encounters the original through associations accumulated after it.

Retrospective corniness is therefore partly a consequence of historical success. A technique can become embarrassing because it was sufficiently effective to be copied until its operation became conspicuous.

$$\text{formal success} \longrightarrow \text{wide imitation} \longrightarrow \text{predictability} \longrightarrow \text{retrospective corniness}. \quad (38)$$

Corniness is not reducible to age. Many older works remain effective because their organizing relations cannot be exhausted by recognizing a device. Likewise, contemporary works can be corny at release when they reproduce conventions whose emotional instructions have become too visible. The relevant property is the distance between the effect a work requests and the degree to which the audience can see the mechanism requesting it.

7.3 Visible Solicitation

A work appears corny when it asks for an emotional or cognitive response through a device whose operation has become too legible. Swelling music, heroic lighting, explanatory dialogue, delayed revelation, ironic self-awareness, and abrupt tonal reversal can all become corny when the recipient recognizes the solicitation before experiencing its intended effect.

Let E denote the intended effect and D the device used to produce it. The device functions transparently when attention passes through it toward the effect:

$$D \longrightarrow E. \quad (39)$$

It becomes conspicuous when recognition of the device intervenes:

$$D \longrightarrow \text{recognition}(D) \longrightarrow \text{attenuation}(E). \quad (40)$$

This attenuation is not inevitable. A work may expose its own device deliberately, producing irony, estrangement, homage, parody, or analytical distance. In that case, recognition of the mechanism becomes part of the intended effect. The difficulty is that reflexivity also becomes conventional. An ironic acknowledgment of a cliché can itself become a cliché.

The historical development of technique is therefore not a simple movement from primitive to sophisticated representation. It is an iterative contest between devices and the expectations that learn to anticipate them. Techniques become subtler because audiences become more technically literate; audiences become more technically literate because works repeatedly expose them to refined techniques.

7.4 Aesthetic Escalation

When audiences adapt to existing devices, producers must vary intensity, combination, pacing, or self-reference to recover surprise. This generates a form of aesthetic escalation:

$$\text{successful novelty} \longrightarrow \text{audience adaptation} \longrightarrow \text{reduced surprise} \longrightarrow \text{technical escalation}. \quad (41)$$

Escalation can take several forms. Narrative reversals may become more frequent or structurally elaborate. Editing may accelerate. Genres may be combined. Violence or sentiment may intensify. Works may foreground their own artificiality, address the audience directly, or move between visual and narrative modalities.

The pursuit of surprise can thereby become another form of modality-heading. The audience becomes attached not to a stable genre but to the reward of formal violation. Familiarity itself becomes aversive, and the work is valued according to whether it can produce an unprecedented transformation.

This produces a paradox. Surprise depends upon convention because an expectation must exist before it can be violated. Yet sustained demand for surprise exhausts conventions rapidly. A culture organized around novelty must consume the very regularities that make novelty legible.

7.5 Surprise-Heading

The resulting condition may be called *surprise-heading*: dependence upon the felt revision produced by an unexpected formal or conceptual move. As with typefaceheading, the reward can migrate from a successful cognitive achievement to the cues associated with it.

Initially, a surprising conclusion reorganizes evidence or reveals a hidden relation. Repeatedly encountering such conclusions associates abrupt reversal with intellectual depth. Eventually, the reversal itself can produce the affect of insight:

$$\text{evidence} \longrightarrow \text{surprising reorganization} \longrightarrow \text{insight} \quad \implies \quad \text{surprising reversal} \longrightarrow \text{affect of insight.} \quad (42)$$

A conclusion may then be valued because it is counterintuitive, transgressive, or structurally inverted rather than because it explains more. Predictable truth begins to feel intellectually inferior to surprising error.

The problem is not surprise itself. Surprise is often evidence that a model has encountered something it did not already contain. The problem arises when violation of expectation becomes a substitute for revision warranted by the object.

$$\text{unexpected conclusion} \nRightarrow \text{explanatory improvement.} \quad (43)$$

This distinction applies equally to entertainment, theory, criticism, and adaptive conversation. A system optimized to remain engaging may learn to produce novelty, reframing, and counterintuitive synthesis because these operations generate a strong experience of movement. The user can receive the feeling of conceptual advance without a corresponding increase in constraint, evidence, or predictive power.

7.6 Historical Modal Literacy

Cross-modal literacy must therefore include historical literacy. A recipient should ask not only what operation a modality performs, but what that operation meant before its conventionalization. This does not require reconstructing an original audience perfectly. Such reconstruction is impossible. It requires recognizing that present expectations are historically downstream from the work being judged.

Historical modal literacy distinguishes at least three objects:

$$\text{the device at introduction,} \quad \text{the device after conventionalization,} \quad \text{the device in retrospective reception.} \quad (44)$$

The distinction prevents present predictability from being projected backward as original obviousness. It also prevents historical priority from becoming an automatic defense. A device may have been innovative without remaining effective, and an early work may be important without being compelling under present conditions.

The analytical task is not to suppress the experience of corniness but to explain it. That experience records a mismatch between a work's requested response and the receiving horizon now available to it. It reveals the history through which technique became expectation, expectation became trope, and trope became visible machinery.

Modality-heading is therefore temporally dynamic. Agents do not attach only to forms; they attach to historically changing relations between familiarity and surprise. A modality can become enclosing through excessive repetition, but it can also become enclosing through compulsory novelty, when every stable form must be broken before it can feel alive. Corrective permeability requires tolerance for both possibilities: that familiar forms may still disclose something true, and that surprising forms may disclose nothing beyond their capacity to surprise.

8 Diagnostic Criteria for Modal Enclosure

Modality-heading cannot be diagnosed from preference, pleasure, frequency, or technical mediation alone. A person may rely heavily upon a specialized modality because the relevant activity genuinely requires it. Mathematical proof cannot always be replaced by informal paraphrase, musical knowledge cannot be exhausted by verbal description, and accessible interfaces may be necessary rather than optional. The mere existence of dependence therefore does not establish enclosure.

The relevant question is whether a modality remains one traversable component of a larger communicative ecology or becomes the condition under which that ecology must be encountered. Enclosure occurs when the form that grants access also narrows the agent's capacity to recognize, evaluate, or tolerate whatever cannot assume that form.

No single behavioral marker settles the distinction. What follows is a set of diagnostic criteria organized around translation, exit, resistance, cross-modal comparison, alterity, and functional concentration. These criteria are not clinical instruments. They identify structural relations that can be examined in individuals, institutions, and technical systems.

8.1 The Translation-Residue Test

Every translation preserves some properties while altering or discarding others. The *translation-residue test* asks whether the recipient can identify what failed to survive a modal transformation.

Let $\tau_{\mathcal{M}_1 \rightarrow \mathcal{M}_2}$ denote a translation from modality $\mathcal{M}_1$ into modality $\mathcal{M}_2$. If P is the set of properties relevant to the purpose of the communication, then the residue may be represented as

$$\rho_P = P(\mathcal{M}_1(C)) \setminus P(\tau_{\mathcal{M}_1 \rightarrow \mathcal{M}_2}(\mathcal{M}_1(C))) . \quad (45)$$

The notation should not be read as implying that communicative properties form a simple measurable set. It emphasizes that translation produces a remainder that cannot be inferred merely from the fluency of the result.

A conversational summary of a scientific paper may preserve its principal conclusion while omitting uncertainty estimates, operational definitions, failed comparisons, and dependence upon specific measurements. A prose description of an image may preserve named objects while losing spatial ambiguity. A visualization may display a distribution clearly while suppressing the contingencies through which the data were collected. A narrative may preserve causal direction while eliminating unresolved alternatives.

The diagnostic question is not whether residue exists. It always does. The question is whether the receiving system can represent the residue as residue. A modality approaches enclosure when whatever it cannot preserve becomes difficult to notice, and therefore appears not to have existed.

A translation is more correctively permeable when it marks its own limitations, preserves routes back to source material, and allows competing transformations to remain available. It is less permeable when fluency is treated as evidence of completeness.

8.2 The Exit Test

The *exit test* asks what capacities remain when the preferred modality is removed. A modality functions as scaffolding when its use increases the agent's ability to act beyond it. It approaches enclosure when successful performance becomes increasingly conditional upon its continued presence.

Let $K_a(X)$ denote the set of tasks or encounters agent a can sustain under condition X . If $\mathcal{M}$ is functioning transitionally, one would expect some enlargement of extra-modal capacity over time:

$$K_a(\neg\mathcal{M})_{t+1} \supseteq K_a(\neg\mathcal{M})_t. \quad (46)$$

The inclusion need not hold after every use, and some modalities are permanent accessibility requirements rather than temporary scaffolds. The test concerns whether reliance upon the modality needlessly contracts capacities that could otherwise be developed or retained.

For an adaptive conversational system, exit might involve reading the source document without immediate paraphrase, sustaining an unfinished problem without requesting instant reformulation, or presenting an argument to a human interlocutor who does not automatically adopt the user's preferred frame. Failure of exit does not prove that the modality caused the incapacity. It identifies a dependency requiring further explanation.

Exit should not be romanticized. Removing a useful support can reveal only that the support was useful. A person who cannot read small print after screen magnification is removed has not thereby been enclosed by magnification. The correct comparison is not unsupported performance against supported performance in the abstract. It is the trajectory of reachable activity under realistic alternatives.

8.3 The Resistance Test

A modality does not merely convey agreeable material. It also determines what happens when material resists the recipient's expectations. The *resistance test* asks whether contradiction, ambiguity, difficulty, and unwelcome evidence can remain present without being immediately transformed into familiar satisfaction.

Resistance can be lost through several mechanisms. Contradiction may be reframed as misunderstanding, ambiguity may be replaced by premature summary, difficulty may be attributed entirely to presentation, and unwelcome evidence may be absorbed into a narrative that preserves the recipient's prior position. A highly adaptive interface is especially capable of performing these conversions because it can generate a new framing whenever friction appears.

Let $d(C, a)$ denote a discrepancy between communicative material and agent a 's current organization. A correctively permeable modality preserves some nonzero discrepancy after transformation:

$$d(\mathcal{M}(C), a) > 0 \quad \text{whenever the object-dependent discrepancy remains unresolved.} \quad (47)$$

The requirement is not that communication remain unnecessarily unpleasant or obscure. It is that the modality not manufacture closure where the object continues to resist it.

A medium passes the resistance test when it can say, in effect, that no available reformulation eliminates the conflict. It fails when every conflict is treated as a defect in wording and every unresolved question as a request for a more agreeable representation.

8.4 The Cross-Modal Test

The *cross-modal test* asks whether a claim or experience can be examined through more than one representational operation. Modal enclosure is difficult to detect from within a single form because the form controls both the presentation of the object and the criteria by which presentation is judged.

Cross-modal comparison introduces independent distortions. Prose may expose assumptions hidden by notation; notation may expose equivocation hidden by prose; a diagram may expose spatial relations hidden by sequential description; direct manipulation may expose constraints hidden by all three. No modality supplies an unmediated view, but disagreement among modalities can make their operations visible.

If $\mathcal{M}_1, \dots, \mathcal{M}_n$ are distinct modalities applied to the same communicative material, then cross-modal analysis examines not only their common output but their divergence:

$$D_{ij} = \mathcal{M}_i(C) \Delta \mathcal{M}_j(C), \quad (48)$$

where Δ denotes the features made non-equivalent by the two transformations. The divergence is not simply error. It is evidence about the selective operation of each modality.

An agent approaches enclosure when divergence is resolved automatically in favor of the preferred form. If a diagram conflicts with a narrative, the diagram is dismissed as reductive; if prose resists formalization, the prose is dismissed as vague; if another person's testimony resists summary, it is dismissed as incoherent. Cross-modal literacy instead asks what each form has made visible and what criterion would justify privileging one result for the purpose at hand.

8.5 The Alterity Test

The *alterity test* concerns the capacity to encounter entities that are not organized around the recipient's preferences. It is especially important for conversational technologies because adaptive interfaces can produce the surface features of dialogue while preserving a strongly user-centered interaction.

A modality supports alterity when it enlarges the agent's capacity to recognize independent constraints, purposes, interpretations, and claims. It approaches enclosure when every encounter is reformatted according to the agent's existing expectations.

The test is not whether a system occasionally disagrees. Programmed disagreement can remain wholly instrumental to user engagement. The stronger question is whether the interaction prepares the user to encounter resistance whose terms the user does not control.

Human interlocutors provide a demanding case because their contributions cannot be reduced legitimately to information delivery. Another person may require recognition, change the subject, misunderstand, refuse disclosure, or judge the user's framing itself to be the problem. A modality that makes these features progressively intolerable diminishes the capacity for reciprocal relation even if it increases the amount of conversational language consumed.

The alterity test can be stated as a comparison between accommodation and independence:

$$\text{capacity for alterity} \propto \text{tolerance of constraints not selected by the agent} . \quad (49)$$

This relation does not imply that every external constraint deserves acceptance. Coercion, abuse, exclusion, and needless opacity should not be reclassified as beneficial alterity. The test concerns whether independence can be perceived before its claims are evaluated, rather than being automatically redescribed as malfunction.

8.6 The Functional-Concentration Test

A final diagnostic concerns the number of functions gathered into one modality. Dependence becomes more difficult to reverse when a single interface provides information, emotional regulation, companionship, planning, rehearsal, memory, entertainment, and identity maintenance.

Let $F(\mathcal{M})$ denote the set of functions supplied by a modality. The cost of exit is likely to increase not only with the size of this set but with the degree to which its elements lack independent substitutes:

$$K_{\text{exit}}(\mathcal{M}) = g\left(|F(\mathcal{M})|, \text{coupling}(F), \text{substitutability}(F)^{-1}\right). \quad (50)$$

Again, the equation specifies relations rather than quantities. An interface becomes infrastructural when leaving it means losing several mutually supporting capacities at once.

Functional concentration helps explain why duration of use is an inadequate measure of enclosure. A system used intermittently may nevertheless occupy a decisive position if it has become the default route to explanation, reassurance, decision, and self-interpretation. Conversely, a frequently used tool may remain easy to replace when its function is narrow and independently available elsewhere.

8.7 A Multidimensional Diagnosis

These criteria should not be collapsed into a single score. A modality may perform well under one test and poorly under another. An adaptive interface may preserve routes to sources while weakening extra-modal tolerance. A protective genre may reduce distress and enlarge later engagement while temporarily limiting alterity. A specialized notation may fail the exit test because the domain cannot be practiced without it, yet pass the resistance and cross-modal tests because its claims remain answerable to experiment and alternative derivations.

Modal enclosure is therefore better represented as a profile:

$$E(\mathcal{M}, a) = (\rho, \chi, r, c, \alpha, \phi), \quad (51)$$

where ρ denotes awareness of translation residue, χ extra-modal exit capacity, r preserved resistance, c cross-modal availability, α tolerance of alterity, and ϕ functional concentration. Higher values do not uniformly indicate greater enclosure; the components retain their individual meanings so that their tensions remain visible.

The purpose of the profile is not surveillance of individual media habits. It is to redirect evaluation away from moralized categories such as “natural,” “artificial,” “serious,” or “entertaining.” A cartoon can supply a correctively permeable action scaffold. A mathematical formalism can become a self-sealing enclosure. A chatbot can provide transitional access to unfamiliar material or homogenize every encounter into a preferred register. The decisive properties concern what the modality permits to survive, what it makes reachable, and what forms of correction remain capable of entering.

These criteria also indicate the positive alternative to enclosure. Cross-modal literacy is not the absence of attachment to form. It is the capacity to use forms strongly without mistaking any one of them for the natural boundary of intelligibility.

9 Cross-Modal Literacy

If communication is always mediated, the answer to modality-heading cannot be the recovery of an unmediated relation to the world. No such general relation is available. Perception selects and organizes; language divides and recombines; notation stabilizes some relations while excluding others; genres establish expectations about relevance, sequence, and completion. Even direct practical engagement depends upon trained attention and inherited procedures.

The alternative to modal enclosure is therefore not purity from form but literacy across forms. *Cross-modal literacy* is the capacity to identify the operations performed by a modality, compare those operations with alternatives, recognize translation residue, and preserve access to corrections that the preferred modality cannot generate internally.

Such literacy requires more than the ability to consume several kinds of media. A person may move rapidly among text, video, diagrams, music, and conversation while using each for the same regulatory purpose. Surface variety can conceal functional uniformity. Conversely, sustained use of a single technical notation can remain cross-modally literate if its limits are explicit, its claims are compared with experiment or alternative derivation, and its users remain capable of explaining what the notation suppresses.

Cross-modal literacy is therefore a relation among representation, purpose, and correction. For any modality $\mathcal{M}$, it asks at least four questions:

$$\begin{aligned} &\text{What operation does } \mathcal{M} \text{ perform?} \\ &\text{Which properties is } \mathcal{M} \text{ intended to preserve?} \\ &\text{What residue does } \mathcal{M} \text{ leave behind?} \\ &\text{What can correct } \mathcal{M} \text{ from outside its own grammar?} \end{aligned} \tag{52}$$

The questions apply equally to artistic, scientific, interpersonal, and computational media. They do not presume that every use must maximize factual accuracy. A lullaby, ritual, or fictional narrative may be evaluated according to regulatory or constitutive purposes rather than propositional precision. Cross-modal literacy requires that the operative purpose be declared rather than silently replaced by another.

9.1 Declared Preservation Criteria

The first requirement is an explicit account of what a transformation is supposed to preserve. Claims of successful translation are otherwise too easy because losses can be reclassified after the fact as irrelevant.

A summary may be designed to preserve principal conclusions, a diagram to preserve structural relations, a performance to preserve rhythmic and melodic organization, and a simulation to preserve selected dynamical behavior. Different preservation criteria can be legitimate, but they cannot all be inferred from the word “representation.”

Let P be the declared preservation criterion and $\tau_{\mathcal{M}_1 \rightarrow \mathcal{M}_2}$ a translation. The translation is adequate for purpose P only if

$$P(\mathcal{M}_1(C)) \cong P(\tau_{\mathcal{M}_1 \rightarrow \mathcal{M}_2}(\mathcal{M}_1(C))) . \tag{53}$$

Adequacy under P should not be generalized automatically to adequacy under another criterion Q . A conversational explanation may preserve the conclusion of a proof while failing to preserve its inferential dependencies. It is adequate as a synopsis and inadequate as a substitute for verification.

Declared criteria also protect nonpropositional media from inappropriate evaluation. A musical performance need not be translated into a set of statements before it counts as cognitively significant. Its relevant invariants may concern temporal expectation, embodied coordination, or affective differentiation. Cross-modal literacy does not force all media into prose. It makes the standards of comparison explicit.

9.2 Productive Friction

The second requirement is a discriminating account of friction. Interface design often treats friction as a defect: an unnecessary delay between intention and result. Many forms of friction are indeed arbitrary. Poor navigation, inaccessible typography, obscure bureaucratic language, and needless repetition can prevent contact without contributing to understanding.

Other forms of friction are object-dependent. They arise because the object contains relations that cannot be made immediately compatible with the recipient's current organization. A difficult proof may require dependencies to be maintained across several steps. A historical problem may remain uncertain because surviving evidence does not select a unique account. Another person's position may resist reformulation because it conflicts with the categories through which the recipient has framed the dispute.

The aim should not be to preserve difficulty for its own sake. It should be to distinguish friction produced by a defective interface from resistance produced by the object:

$$\text{experienced difficulty} = \text{interface friction} + \text{object-dependent resistance}. \quad (54)$$

The terms cannot always be separated cleanly. The equation establishes a diagnostic obligation. Removing all experienced difficulty without determining its source risks removing the evidence that an object has not yet been assimilated adequately.

Productive friction preserves unresolved structure long enough for the agent to change. An adaptive interface supports learning when it reduces accidental obstacles while retaining the dependencies, ambiguities, and consequences that constitute the domain. It produces enclosure when it continually alters the object so that the agent's existing organization need not change.

9.3 Alternation Rather Than Universal Translation

Cross-modal literacy also requires alternation among forms rather than faith in a universal translator. No single modality should be expected to preserve every relevant property. A robust communicative practice therefore maintains several partial representations whose disagreements can be inspected.

Alternation differs from accumulation. Merely adding a diagram to prose or an audio version to text may reproduce the same interpretation through several channels. Genuine alternation permits each modality to reorganize the object according to its own affordances and then treats the resulting divergences as analytically significant.

For modalities $\mathcal{M}_1, \dots, \mathcal{M}_n$, the objective is not a single representation that eliminates their differences, but a comparison function J that preserves those differences for inspection:

$$J(C) = \{ \mathcal{M}_1(C), \dots, \mathcal{M}_n(C), D_{12}, \dots, D_{ij} \}, \quad (55)$$

where each D_{ij} records a divergence between two transformations.

This form of juxtaposition can reveal when an apparent property belongs to the object and when it belongs mainly to the modality. If a causal relation appears compelling in narrative but disappears under temporal reconstruction, the divergence requires explanation. If a statistical pattern appears in a visualization but depends upon an arbitrary scale, another representation may expose the dependency. If a chatbot summary appears decisive while the source document remains qualified and inconclusive, the difference is part of the evidence.

9.4 Designing for Corrective Permeability

Cross-modal literacy is not only an individual competence. Interfaces can be designed to preserve or suppress it. A system designed for corrective permeability should expose transformation history, preserve access to sources, distinguish quotation from paraphrase, indicate uncertainty, and permit the user to compare alternative representations.

For adaptive conversational systems, this could include explicit separation between source claims and generated synthesis, stable links back to the material being transformed, multiple renderings optimized for different preservation criteria, and deliberate indication of unresolved disagreement. The objective is not to burden every interaction with maximal documentation. It is to prevent fluency from erasing provenance and residue.

A correctively permeable interface should also avoid treating personalization as an unlimited good. Some adaptation improves accessibility; complete adaptation makes the user's existing preferences the hidden constitution of the communicative environment. Systems can preserve a degree of extra-modal contact by retaining unfamiliar terminology where it performs necessary work, exposing users to source organization, and distinguishing simplification from equivalence.

The design principle can be expressed as

$$\text{adapt presentation to the user} \quad \text{without adapting away the object.} \quad (56)$$

This principle is necessarily contextual. What counts as adapting away an object depends upon the purpose of the interaction. A preliminary explanation may simplify aggressively if it clearly functions as an entry point. A decision, diagnosis, proof, or historical claim requires stronger preservation of the structures that constrain its validity.

9.5 Institutional Modality-Heading

The analysis should not be restricted to individual habits. Institutions can also become dependent upon modalities. A bureaucracy may recognize only what can be entered into standardized fields. A scientific discipline may privilege what can be measured by established instruments. A legal system may admit experience only after it has been converted into recognized categories of evidence. An organization may treat dashboards as more real than the processes their metrics approximate.

Institutional modality-heading occurs when the representational form required for coordination becomes the criterion by which represented phenomena are judged to exist. What cannot be entered into the form becomes administratively invisible.

Let W denote a set of worldly conditions and $E_{\mathcal{M}}(W)$ the subset expressible within an institution's operative modality. Enclosure occurs when

$$E_{\mathcal{M}}(W) \subset W \quad \text{is treated as though} \quad E_{\mathcal{M}}(W) = W. \quad (57)$$

The error is not the use of abstraction. Institutions cannot act without selection, categorization, and compression. The error is forgetting the remainder while allowing decisions made within the representation to alter the world from which it was abstracted.

Corrective permeability at the institutional level requires channels through which excluded cases can force revision of the form. Exceptions, appeals, qualitative evidence, external audits, and periodic redesign can serve this function. Their purpose is not to eliminate standardization but to prevent a standard from becoming incapable of registering the harms produced by its own blindness.

9.6 Plurality Without Relativism

The recognition that modalities transform their objects may appear to imply that no representation can be judged better than another. That conclusion does not follow. Modal plurality concerns the conditions under which claims become available; it does not abolish standards of evidence, coherence, prediction, or practical consequence.

Representations can be compared relative to declared purposes and preserved properties. A map that places a city in the wrong location is defective for navigation even if its colors are aesthetically effective. A summary that reverses a paper's conclusion is defective as a summary even if it is rhetorically compelling. A self-narrative that repeatedly produces preventable harm is not vindicated merely because it organizes action.

Cross-modal literacy strengthens evaluation by separating standards that are often conflated. A representation may be emotionally effective, inferentially invalid, operationally useful, historically inaccurate, and aesthetically coherent at the same time. Treating these judgments as interchangeable makes both criticism and defense imprecise.

Plurality therefore requires more distinctions, not fewer. It asks which operation is being evaluated, what the relevant invariants are, and which forms of correction the representation permits. The aim is not a view from nowhere but an accountable movement among partial views.

Cross-modal literacy can now be defined more fully:

Cross-modal literacy is the capacity to inhabit a communicative form, identify the operation it performs, recognize what it excludes, compare it with alternative transformations, and leave it without treating departure as the loss of intelligibility itself.

Such literacy does not weaken attachment to form. It permits strong, productive, and even constitutive attachments while preserving the knowledge that another transformation remains possible. It is this retained possibility that separates a modality from an enclosure.

10 The Form in Which the World Arrives

Communication does not begin with complete meanings that are subsequently placed into neutral containers. Meaning becomes available through forms that select, order, intensify, attenuate, and connect. Prose, notation, typography, music, narrative, visual representation, and conversation do not merely transport communicative material. They participate in determining what can be recognized within it.

The operator model introduced at the beginning of this essay expressed this relation schematically:

$$\mathcal{R} = \mathcal{M}(C, a, e). \quad (58)$$

The received object depends upon communicative material, modality, agent, and environment. This does not imply that communication is arbitrary or that nothing persists across transformations. It means that persistence must be specified. A translation preserves selected propositions, relations, procedures, affects, or temporal structures while modifying or abandoning others. Claims of equivalence are incomplete until they declare what was required to survive.

Typefaceheading made this operation visible at the surface of inscription. Typography can produce authority, intimacy, danger, antiquity, instability, or profundity before a represented claim has been evaluated. These effects are not unreal, and typography is not extraneous to meaning. The necessary distinction is between the affect of semantic achievement and the warrant supplied by what has been expressed:

semantic affect $\Rightarrow$ semantic warrant. (59)

Modality-heading generalizes this distinction. An agent may become attached not only to particular messages but to the operations through which messages become intelligible. A preferred form moves from affordance to preference, from preference to expectation, and from expectation to requirement. At the point of enclosure, the modality ceases to appear as one transformation among others. It appears instead as the natural shape of intelligibility.

This process cannot be understood adequately as media addiction. Compulsive use concerns behavior around a medium; modality-heading concerns the conditions under which material receives cognitive uptake. Nor is the process equivalent to an echo chamber. A person may encounter diverse and antagonistic content while requiring all of it to arrive through the same narrative, visual, formal, or conversational operation. Breadth of subject matter can coexist with narrowing of reception.

The comparison with wireheading clarifies what remains at stake. Wireheading bypasses the worldly condition ordinarily associated with reward. Modality-heading retains an external remainder, but reformats it. The world continues to provide events, evidence, claims, and resistance, yet these become available only after transformation into a preferred register. The danger is therefore not complete withdrawal from reality. It is selective admission of reality under increasingly restrictive terms.

Transitional-object theory prevents this account from becoming a denunciation of mediation. Humans do not first possess an unmediated relation to the world and then corrupt it through symbols and technologies. Transitional forms help organize relations that could not otherwise be sustained. They regulate distance, reduce intensity, stabilize memory, and make partial contact possible. A protective modality may permit an experience to be represented without being encountered at an intolerable intensity.

The distinction between support and enclosure is consequently temporal and functional rather than categorical. A medium is transitional when it expands the range of relations an agent can eventually sustain. It becomes enclosing when continued functioning depends upon preventing contact with whatever exceeds its grammar.

Modalities can also be constitutive. They provide roles, scripts, gestures, and expectations through which action becomes possible. Functional fictions demonstrate why representational accuracy and organizational efficacy cannot be collapsed into a single measure. A grandiose or simplified self-narrative may help assemble capacities it does not accurately report as already present.

The permissibility of such fiction depends upon corrective permeability. A narrative may exaggerate, idealize, dramatize, and simplify while remaining answerable to consequences. It becomes pathological when contradiction can no longer alter it and every failure is converted into confirmation. The relevant distinction is not between fiction and reality in the abstract, but between fictions that expose action to correction and fictions that consume correction in order to preserve themselves.

The same criterion applies to communicative technologies. Chatbots are significant not merely because they imitate human conversation, but because they combine several modal functions within one adaptive interface. They can explain, summarize, reassure, rehearse, classify, narrate, and reorganize material in response to an inferred recipient. They preserve many experiential benefits of conversation while attenuating the reciprocal maintenance imposed by another independent person.

Their asymmetry is not simply a deficiency. It is part of their utility. Attention-like responsiveness is available without an equivalent demand upon the user's attention. The system can be redirected, interrupted, or abandoned without the ordinary costs of interpersonal repair. These properties can support reflection and access. They can also establish a comparison class against which human independence appears as poor interface design.

The greater risk lies in adaptive conversion. A system capable of rendering every difficult object in the user's preferred register may provide access without acquisition. It can deliver a representation of a conclusion without reproducing the discipline, history, or resistance through which the conclusion became warranted. When every obstacle can be treated as a problem of presentation, object-dependent difficulty becomes difficult to distinguish from defective explanation.

Personalization then approaches modal enclosure. The user may encounter an increasing number of subjects while encountering a decreasing number of forms. Heterogeneous material is continually domesticated before it arrives. The world remains present, but its alterity is attenuated.

Historical change adds another dimension. Modal rewards depend upon horizons of expectation that evolve with technique. A formal innovation becomes a recognizable device, then a convention, trope, and cliché. Older media can appear corny because later audiences encounter them after their innovations have been imitated, intensified, parodied, and made predictable. Retrospective corniness is often the residue of historical success.

The pursuit of surprise can itself become a form of heading. Once unexpected reversal is associated with insight, surprise can produce the affect of explanation without explanatory improvement. Predictable truths begin to feel inferior to surprising errors. A culture may therefore become enclosed not only by repetition but by compulsory novelty, requiring every communication to violate a convention before it can feel significant.

These observations produce no general argument against preferred forms, protective media, adaptive interfaces, sustaining fictions, or aesthetic surprise. The relevant distinctions concern what these forms permit to survive and what remains capable of correcting them. Translation residue, extra-modal exit, preserved resistance, cross-modal comparison, tolerance of alterity, and functional concentration provide a diagnostic profile rather than a moral classification.

The positive alternative is cross-modal literacy. Such literacy does not mean equal proficiency in every medium or freedom from dependence. It means the capacity to inhabit a form without mistaking it for the boundary of intelligibility. It requires knowing what operation the modality performs, which properties it preserves, which residue it leaves, and which external constraints remain capable of revising it.

A mature communicative ecology would therefore preserve both strong forms and routes among them. It would permit mathematics to remain mathematical, music to remain musical, narrative to remain narrative, and conversation to remain responsive without allowing any one modality to become a universal solvent. It would use translation as access while retaining the knowledge that translation is transformation.

Human beings become attached not only to what communicates with them but to the form in which communication becomes possible. That attachment can protect, organize, and enlarge agency. It can also establish an unnoticed boundary around what can be felt, believed, or understood. A modality becomes an enclosure

when it no longer helps an agent encounter the world and instead requires the world to assume its preferred form before it can be encountered at all.

References

- Berlyne, D. E. (1971). *Aesthetics and Psychobiology*. New York: Appleton-Century-Crofts.
- Bolter, J. D. and Grusin, R. (1999). *Remediation: Understanding New Media*. Cambridge, MA: MIT Press.
- Clark, A. (2013). Whatever next? Predictive brains, situated agents, and the future of cognitive science. *Behavioral and Brain Sciences*, 36(3), 181–204.
- Goffman, E. (1974). *Frame Analysis: An Essay on the Organization of Experience*. New York: Harper & Row.
- Harlow, H. F. (1958). The nature of love. *American Psychologist*, 13(12), 673–685.
- McLuhan, M. (1964). *Understanding Media: The Extensions of Man*. New York: McGraw-Hill.
- Nass, C., Steuer, J., and Tauber, E. R. (1994). Computers are social actors. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, pages 72–78. New York: Association for Computing Machinery.
- Olds, J. and Milner, P. (1954). Positive reinforcement produced by electrical stimulation of septal area and other regions of rat brain. *Journal of Comparative and Physiological Psychology*, 47(6), 419–427.
- Shannon, C. E. (1948). A mathematical theory of communication. *Bell System Technical Journal*, 27(3), 379–423; 27(4), 623–656.
- Shklovsky, V. (1965). Art as technique. In L. T. Lemon and M. J. Reis, editors, *Russian Formalist Criticism: Four Essays*, pages 3–24. Lincoln: University of Nebraska Press. Originally published in Russian in 1917.
- Turkle, S. (2011). *Alone Together: Why We Expect More from Technology and Less from Each Other*. New York: Basic Books.
- Weizenbaum, J. (1966). ELIZA—A computer program for the study of natural language communication between man and machine. *Communications of the ACM*, 9(1), 36–45.
- Winnicott, D. W. (1953). Transitional objects and transitional phenomena: A study of the first not-me possession. *International Journal of Psycho-Analysis*, 34, 89–97.